Akash Raj

UG (III Year I Semester)
B.Tech. (Mechanical Engineering)
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Indian Institute of Technology Roorkee

VERIFIED

Area of Interest

Machine Learning, Data Science, Competitive Programming, Software Engineering

Education

Year	Degree/Examination	Institution/Board	CGPA/ Percentage
2024	B.Tech. 2nd Year	Indian Institute of Technology, Roorkee	8.449
2023	Intermediate (Class XII)	65140 S D S VIDYA MANDIR PO UDAYPUR SAMASTIPUR BIHAR (CBSE)	94.40 %
2021	Matriculate (Class X)	10912-DECENT SCHOOL RAJNAGAR KOTA RAJASTHAN (CBSE)	90.40 %

Internships

SWE Co Op Intern | AlgoUniversity — May 2025 - July 2025

- Launched an online judge platform allowing users to submit and evaluate code solutions in <500ms.
- Designed and engineered a user-friendly interface to enhance user experience.
- Programmed robust backend services using Django for user authentication, problem submission and code evaluation.
- Integrated a secure code compiler enabling execution of 1000+ isolated user submitted code, ensuring security and
- reliability and Containerized the application using Docker to ensure consistency and scalability across environments.
- Managed data operations and deployed the platform on AWS EC2.
- Received technical mentorship from engineers at Google London, Apple, ByteDance Singapore, and Alphagrep Singapore

Projects

Comparative Analysis of European Option Pricing | Finance Club, IIT Roorkee — May 2025 - June 2025

- Implemented European option pricing models using Binomial Tree (with variable time steps) and Black-Scholes
- formula(also computes option greeks) .Binomial tree gives same result(5 decimal places) as BS after 200 steps.
- Analysed sensitivity across key parameters (volatility (10-60%), time to maturity(1 week to 1 year), interest rate, strike
- price).Validated model predictions with real data (AAPL, SPY, NIFTY50) achieving less than 10% error for T=1Y.
- Reported accuracy,assumptions, computational efficiency. Visulized results (12+plots) using Matplotlib/Seaborn.

Automatic Modulation Classification model for 6G Short-Range Underwater Acoustic Communication(UAC) | Prof. Sneha Singh, MIED, IIT Roorkee — May 2025 - July 2025

- Student assistant in this project which is a subproject of 6G short Range UAC which utilizes 5L FC DNN with MAML.
- Used open source datasets(1000+ samples) and simulated own signal data using NumPy, matplotlib to map noisy signals
- to their corresponding clean signals.Developed an Automatic Modulation Classifier model using Convolution neural
- networks. The AMC is trained to detect 3+ modulation techniques including BFSK, BPSK, OFDM etc.
- Presented the project idea and solution at TTDF symposium organized at IIT Madras Research Park

Design of path following robot | IIT Roorkee — July 2023 - November 2023

- Built a path-following robot using Ultraviolet (UV) sensors to detect lines and make real time navigation decisions.
- Contributed majorly on coding the Arduino based control system, integrating sensor feedback with motor control for smooth and accurate line tracking.

Automated Meta Data Generation | MaRS (Models and Robotics Section), IIT Roorkee — June 2025

- Created an automated metadata generation system for various unstructured document types like pdf, docx, txt, capable of extracting text content from diverse formats incorporating optical character recognition (OCR).
- Supports 3+ input file formats (.pdf, .docx and image (.jpg, .png , .jpeg)), converts to image during processing using openCV (rescaling, denoising, border handling), pytesseract, docx, pymupdf, pdf2image.
- Web interface for document upload and metadata viewing 8 key fields (Title, keywords, Date, No. of Pages, Type of document, summary, No. of characters, No. of words.) using Django.

Skills

Computer languages: C++, Python, Javascript
Software Packages: NumPy,Pandas, Scikit-learn,Version-control(git), Kali Linux, TensorFlow, Django, AWS, SQL

Positions of Responsibility & Extra Curriculars

Volunteer | Web Development Cell, NSS IIT Roorkee — October 2023 - May 2024

- Volunteered in National Social Summit 2024.

References

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